

2nd Year 1st Semester Examination-2020

Department: Computer Science & Engineering

Subject: Object Oriented Programming Language

Total Marks: 70

Course Code: CSE-301

Time: 2 (two) hours

Answer any 03 (three) questions.

- 01 a) Which are the best features of OOPs and why explain? 6
b) What are the advantages of OOP over structural programming? 6
c) Is encapsulation and data hiding same? 6
d) Assign and print the roll number, phone number and address of two students having names "X" and "Y" respectively by creating two objects of class "Student". 5.33
- 02 a) Define copy constructor. 3.33
b) Distinction between static and non-static member. 4
c) What are the uses of access modifiers in C++? Explain through an OOP program. 8
d) Can we have both default constructor and parameterized constructor in the same class? Explain through an OOP program. 8
- 03 a) What is an exception? Discuss how try-catch-finally blocks works in java. 4
b) What is exception handling? Using exception handling mechanism write a program to manage Arithmetic exception in java. 5
c) What happens if an exception is not caught? What will be the output of the following program 5

```
Class A{
    Public static void main(String[] args){
        Try{
            int [] a = {5,10,15};
            System.out.println(a[5]);
            System.out.println(10/0);
            catch(ArithmeticException e){
                System.out.println("Arithmetic Exception");
            }
            catch(ArrayIndexOutOfBoundsException e){
                System.out.println("Hello world");
            }
            System.out.println("Exception Handling");
        }
    }
}
```

- d) What is inline function? Write down the advantages and limitations of inline function. 4.33
e) What is template? What would be the output of the following program : 5

```
#include<iostream>
using namespace std;
template<typename T>
void print_data(T output)
{
    cout<<output<<endl;
}
int main()
```

```

{
    double d = 5.5;
    int a = 10;
    print_data(d);
    print_data(a);
    return 0;
}

```

- 04 a) What is new and delete operator? What are their advantages? 4
- b) Define multiple inheritance. Give an example of ambiguity in multiple inheritance. 4
- c) Explain the concept of operator overloading. What are the ways to overload an operator in c++? 6
- d) How to prevent overriding and inheritance using final keyword? Explain with example. 4
- e) Define static keyword. Write down the differences among public, private and protected access specifier in a class. 5.33
- 05 a) What is the difference between overloading and overriding concepts in C++? Explain the usage of these concepts with suitable example code in C++/Java? 8
- b) How to create a thread? Mention the advantages of multithreading. 6
- c) What will be the output of the following program? 5
1. abstract class Bike{
 2. Bike(){System.out.println("bike is created");}
 3. abstract void run();
 4. void changeGear(){System.out.println("gear changed");}
 5. }
 - 6.
 7. class Honda extends Bike{
 8. void run(){System.out.println("running safely..");}
 9. }
- ```

class TestAbstraction2{
public static void main(String args[]){
 Bike obj = new Honda();
 obj.run();
 obj.changeGear();
}
}

```
- d) Show the life cycle of a thread. 4.33

University of Dhaka  
Affiliated Engineering Colleges  
Department of Computer Science & Engineering  
2<sup>nd</sup> Year 1<sup>st</sup> Semester Examination-2020 (Session: 2018-19)

Subject: Data Structure  
Total Marks: 70

Course Code: CSE-303  
Time: 2 (two) hours

Answer any three (3) questions

- 01 a) Consider the following stack of characters where STACK is allocated N=8 memories: STACK: 5  
A,C,D,F,K, \_ \_ \_  
i. POP(STACK, ITEM).  
ii. POP(STACK, ITEM).  
iii. PUSH(STACK, L).  
iv. PUSH(STACK, P).  
v. POP(STACK, ITEM).
- b) Explain stack and queue with example. Write down an algorithm which PUSHes an item 3+4=7  
onto a stack.
- c) Using algorithms translate to postfix expression and evaluate the following infix expression: 6  
 $P=12/(7-3)+2*(1+5)$
- d) Define Heap. Write down the properties of max-heap and min-heap with example. 1.33+4  
= 5.33
- 02 a) How a 2D array represented in memory. 3  
b) What is Linked list? How linked list overcome the limitation of array. Write down the 2+3+5  
algorithm to implement the following operations of linked\_list =10  
i. to insert an element after the given elements of the list  
ii. to delete an element from the end of the list.
- c) How many number of swapping needed to sort the characters PEOPLE in ascending order, 5  
using BUBBLE SORT algorithm.
- d) Suppose a 10 element array A contain the values a1, a2, ..... a10. Find the values 5.33  
in A after the loop.

Repeat for K = 1 to 9

Set  $A[K + 1] := A[K]$

[End of Loop]

- 03 a) Figure show a linked list in memory. If start=4 and avail=3, write the sequence of characters 5  
in the list. Draw the Linked List after inserting G at the beginning of the list.

|   | INFO | LINK |
|---|------|------|
| 1 | A    | 2    |
| 2 | B    | 8    |
| 3 |      | 6    |
| 4 | C    | 7    |
| 5 | D    | 0    |
| 6 |      | 0    |
| 7 | E    | 1    |
| 8 | F    | 5    |

- b) Suppose LIST is a Linked List in memory. Write an algorithm which deletes the last node 6  
from LIST.
- c) Write down the advantages and disadvantages of array over linked list. 4
- d) Explain with appropriate figure how a node N between Node A and Node B can be deleted 8.33  
from a Linked List?

04 a) Define the following terms with an example:

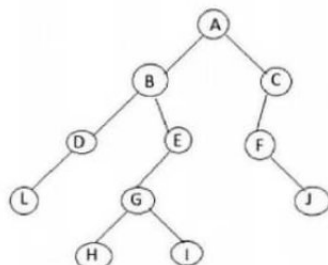
3

- i. Tree
- ii. Full Binary Tree
- iii. Complete Binary Tree

b) From the following Binary Tree find the sequence of nodes When traversing:

6

- i. Pre-Order
- ii. In-Order
- iii. Post-Order

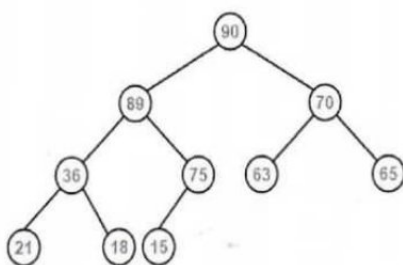


c) Construct a binary search tree inserting the following numbers in order into an empty binary search tree: 40, 60, 20, 50, 55, 35.

6.33

d) Consider the heap following heap. Write down the steps to delete the root of that heap with figures.

8



05 a) Define the following:

6

- i) Multi-Graph ii) Degree of Directed Graph iii) Weighted Graph

b) The following is the adjacency matrix A of an undirected graph G. Draw the graph G and find its incidence matrix M.

3+3 = 6

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$

c) Explain the process how a node can be deleted from a graph.

6.33

d) Write down the Breadth-First Search Algorithm that executes on a graph G beginning at a starting node A.

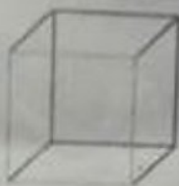
5

Marks: 70

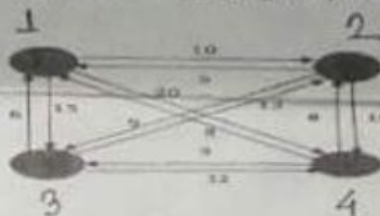
[Answer any five questions. Assume reasonable value for any missing data]

Time: 1.00 Hours

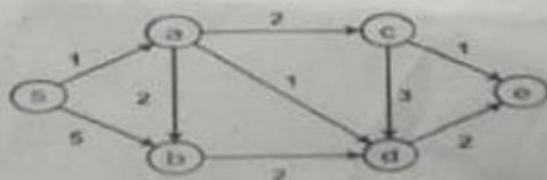
1. a) What is Backtracking? Write down The general steps of backtracking. 4
- b) Write down the Characteristics of Huffman Coding. 4
- c) Draw Six Hamiltonian cycle from the Graph which given below. 6



2. a) Define greedy method. Write down the basic algorithm for greedy approach. 4
- b) What is Travelling Sales Person (TSP)? Solve the following TSP problem for the given matrix and graph (assume root is 1): 5



- c) Write a greedy algorithm for single source shortest path problem. Based on this algorithm find the single source shortest path for the following graph: 5



3. a) Consider the following 9 activities with their starting and finishing time. 7

Activity: (A<sub>1</sub>, A<sub>2</sub>, A<sub>3</sub>, A<sub>4</sub>, A<sub>5</sub>, A<sub>6</sub>, A<sub>7</sub>, A<sub>8</sub>, A<sub>9</sub>)

Start: (1, 2, 3, 4, 5, 6, 7, 8, 9)

Finish: (3, 5, 7, 8, 9, 11, 14, 16)

Compute a schedule where the greatest number of activities takes place Using Activity selection Problem.

- b) Write down the Real-life Applications of Activity Selection Problem. 3
- c) Write down the GREEDY ACTIVITY SELECTOR Algorithm. 4

4. a) Define and explain i) DAG ii) indegree iii) outdegree. 3
- b) What is parallel algorithm? Explain Flynn's Classification of computers with required Diagram. 6
- c) Write down the steps of Interpolation search. 2
- d) Write down the pseudo-code of Graph coloring Problem. 3

A<sub>1</sub>, A<sub>2</sub>, A<sub>3</sub>, A<sub>4</sub>, A<sub>5</sub>

[Answer any five questions. Assume reasonable value for any missing data.]

Time: 3.00 Hours

- a) Define Orthogonal Matrix and Inverse Matrix. Show that every square matrix can be expressed as sum of the symmetric and Skew-symmetric matrix. 5
- b) If  $AB = BA$  then prove that  $(AB)^n = A^n B^n$  6

- c) Prove that  $\frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$  is unitary. 3

- a) Define matrix polynomial. Let  $A = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$  and  $f(\lambda) = \lambda^2 - 4\lambda + 3$  then prove that  $A$  is a root of  $f(\lambda)$ . 3

- b) State and prove Cayley-Hamilton theorem. 5

- c) Verify Cayley-Hamilton theorem for the matrix  $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix}$  6

- a) What is curl of a vector? Prove that  $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ . 4

- b) Find the equation of the tangent plane and normal line to the surface  $z = x^2 + y^2$  at the point  $(2, -1, 5)$ . 6

- c) Find the directional derivative of  $\phi(x, y, z) = 4xz^2 - 3x^2y^2z$  at the point  $(2, -1, 2)$  in the direction  $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$ . 4

- a) Define Fourier series. If  $f(x)$  is an even function then show that (i)  $a_n = \frac{2}{\pi} \int_0^\pi f(x) \cos nx dx$  4

$$(ii) a_n = \frac{2}{\pi} \int_0^\pi f(x) \cos nx dx$$

- b) Find the Fourier series expansion of the periodic function  $f(x) = -x$  when  $-\pi < x < 0$  and  $f(x) = x$  when  $0 < x < \pi$ . Also prove that  $\frac{\pi}{8} = 1 - \frac{1}{3^2} + \frac{1}{5^2} - \dots$  10

- a) Define vector space and subspace. Show that  $S = \{(a, b, c, d) \in \mathbb{R}^4 : 2a - 3b + 5c - d = 0\}$  is a subspace of  $\mathbb{R}^4$ . 5

- b) Define linear dependence and independence of vector. Show that  $\{(1, 1, -1), (1, 0, 2), (1, 1, 1)\}$  vectors are linearly independence. 6

- c) Show that a single vector is linearly dependent and only if it is a zero vector. 3

- a) Solve the initial value problem by Laplace transform  $Y''' - 3Y'' + 3Y' - Y = t^2 e^t$ ;  $Y(0) = 1, Y'(0) = 0, Y''(0) = -2$  5

- b) Using convolution theorem evaluate  $L^{-1} \left\{ \frac{1}{s^2(s+1)^2} \right\}$  5

- c) If  $L\{F(t)\} = f(s)$  then show that,  $L\left\{\int_0^t F(t) dt\right\} = \frac{f(s)}{s}$  4

- a) Show that  $\mathbf{E} = (x^2 - yz)\mathbf{i} + (y^2 - zx)\mathbf{j} + (z^2 - xy)\mathbf{k}$  is a conservative force field. 2

- b) Show that  $\iint_S \mathbf{A} \cdot \mathbf{n} ds = 24$  where  $\mathbf{A} = 18xz\mathbf{i} - 12z\mathbf{j} + 3y\mathbf{k}$  and  $S$  is the part of the plane  $2x + 3y + 6z = 12$  which is located in the first octant. 5

- c) Find the Fourier transform of  $F(x) = \begin{cases} 1-x^2 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$  Also prove that  $\int_0^\pi \frac{(\sin x - x \cos x)}{x^3} \cdot \cos \frac{x}{2} dx = \frac{3\pi}{16}$  7

Marks: 70

[Answer any five questions. Assume reasonable value for any missing data.]

Time: 3.00 Hours

1. a) What is Transformer? Why frequency of primary and secondary of a transformer are same? 5  
 b) Why Transformer rated at KVA? 2  
 c) Explain Transformer name plate Data. 3  
 d) Find the all day efficiency of a 50 KVA distribution transformer having full load efficiency of 94% and full load copper loss equal to constant iron losses. The loading p.f 1.0 is as (i) No load at 10 hours; (ii) half load at 5 hours; (iii) 25% load at 6 hours; (iv) full load at 3 hours. 4
2. a) Why do we need a starter to start a DC motor? 3  
 b) How can we control the speed of stepper motor? 3  
 c) Write about Permanent magnet Brushless AC Generator with circuit Diagram. 4  
 d) A 3 phase induction motor is wound for 4 poles and is supplied from 50-Hz system. Calculate (i) the synchronous speed (ii) The rotor speed when slip is 4% and (iii) Rotor frequency when rotor runs at 600 rpm. 4
3. a) What are the desirable characteristics of a transducer elements? 4  
 b) State the difference between analog and digital indicators. 3  
 c) Explain the operating principle of a ramp type DVM. 4  
 d) A capacitive transducer consists of two plates of diameter 2 cm each, separated by an air gap of 0.25 mm. Find the displacement sensitivity. 3
4. a) Classify DC generator. 3  
 b) What is back e.m.f? Write down the significance of back e.m.f. 3  
 c) A 25-kW, 250-V d.c shunt generator has armature and field resistance of 0.06Ω and 100 Ω respectively. Determine the total armature power developed when working (i) as a generator delivering 25kW output and (ii) As a motor taking 25kW input. 5  
 d) Why single phase induction motor is not self-starting? 3
5. a) Classify digital voltmeters. 2  
 b) What is instrumentation amplifier? Write down the key features of instrumentation amplifier. 4  
 c) Explain the construction and operation of a LED display. 6  
 d) Compare LED and LCD displays. 2
6. a) What are the key features of instrumentation amplifiers? 2  
 b) Explain the construction and operation of a Piezo-electric transducer. 4  
 c) Explain the working principle of a Thermo-electric generator. 5  
 d) Discuss about functional elements of a measurements system. 3
7. a) What do you understand by DAS? 2  
 b) Discuss the operation of a large-scale DAS with required block diagram. 6  
 c) Show the schematic block diagram of a generalized data acquisition system. 4  
 d) State the objectives of a DAS. 2

Marks: 70

[Answer any five questions. Assume reasonable value for any missing data.]

Time: 3.00 Hours

- |                                                                                             |   |
|---------------------------------------------------------------------------------------------|---|
| 2. a) What is diode? Implement NOR gate by diode.                                           | 5 |
| b) Show that Transistor acts as a switch.                                                   | 4 |
| c) Describe Exclusive NOR gate by transistor.                                               | 5 |
| 3. a) What do mean by LED and LCD.                                                          | 4 |
| b) Explain Sample and Hold (S/H) circuit by using Operational Amplifier.                    | 5 |
| c) Describe Analog to digital Converter circuit.                                            | 5 |
| 4. a) What is PLA? Explain block diagram of PLA circuit.                                    | 4 |
| b) Describe the pulse transformer circuit and its equivalent circuit.                       | 5 |
| c) Explain pulse transmission with input and output waveforms in different media.           | 5 |
| 5. a) What do you mean by flip flops? Explain operating principle of J-K flip flop.         | 4 |
| b) Draw & explain BCD ripple counter.                                                       | 5 |
| c) What is the difference between a serial and parallel transfer?                           | 2 |
| d) Explain operating principle of 3-bit register.                                           | 3 |
| 6. a) What do you mean by A/D & D/A converters?                                             | 2 |
| b) Draw the Transistor Transistor Logic (TTL) circuit and explain working procedure.        | 5 |
| c) Explain operating principle of comparator circuit.                                       | 4 |
| d) Write down a comparison between A/D & D/A converters.                                    | 3 |
| 7. a) What do you mean by clippers & clampers circuit?                                      | 2 |
| b) Describe Negative clipping circuit with output waveforms.                                | 5 |
| c) What is Schmitt trigger circuit? Explain operating principle of Schmitt trigger circuit. | 5 |
| d) Write down application of clippers circuit.                                              | 2 |
| 8. a) What do you mean by multivibrators?                                                   | 2 |
| b) Explain operating principle of astable multivibrators.                                   | 6 |
| c) Write down application of monostable multivibrators.                                     | 2 |
| d) Briefly explain various methods of generating a time-base waveform.                      | 4 |